ACT-Tracks with Auxiliary Motion Learning

Native ACT action generation, regularized by past completion and future prediction

Native ACT · Track conditioning · Training only

ACT latent z
posterior at train; 0 at test

$q\varphi(z \mid q_o, a^*)$
Expert actions enter the CVAE only during training.

Shared observation modules

RGB history
3 cameras · 16 frames

Shared ResNet-18
Current: full image grid
Past: 4 × 4 pooled
+ spatial / time encoding

ACT encoder
Visual tokens
+ current joints + latent
Native Transformer

Gated track fusion
Q: memory K,V: tracks
Residual cross-attention
M′ = M + tanh(g) Attn

$M\pi$

ACT decoder
100 action queries
Native Transformer
+ linear action head

Action chunk
100 × 12 joint targets

Current robot state
12 measured joints

M_o

3D track history
16 observed points

Oracle vertex histories

Track MLP
Anchor + displacement
+ time + validity
One token per point

Track tokens P

Shared weights; policy and auxiliary are separate forward passes.

Training-only auxiliary heads

Observation-only pass: z = 0

Past completion
XYZ + past-time query

$\Delta\hat{x}_j(\tau)$ · $\tau < 0$ → ***Lpast***

Future prediction
XYZ + future-time query

$\Delta\hat{x}_j(\tau)$ · $\tau > 0$ → ***Lfuture***

Query anchors: 48 withheld points
Current XYZ only
No auxiliary output enters the actor.

Withheld GT trajectories

Training objectives

$L_{policy} = \text{masked } L1(\hat{a}, a^*) + 10\,KL$

$L_{aux} = 0.5\,L_{past} + 0.5\,L_{future}$

Auxiliary gradients regularize shared modules; shared norm cap = 0.1 × policy norm.

Inference uses only observations → ACT encoder / track fusion → ACT decoder → actions.

Schematics; not measured trajectories